



Today's agenda

↳ Decorator design Pattern

↳ Flyweight design Pattern

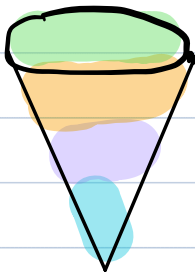


AlgoPrep



* Decorator design Pattern

→ Anal of ICE-Cream management System.

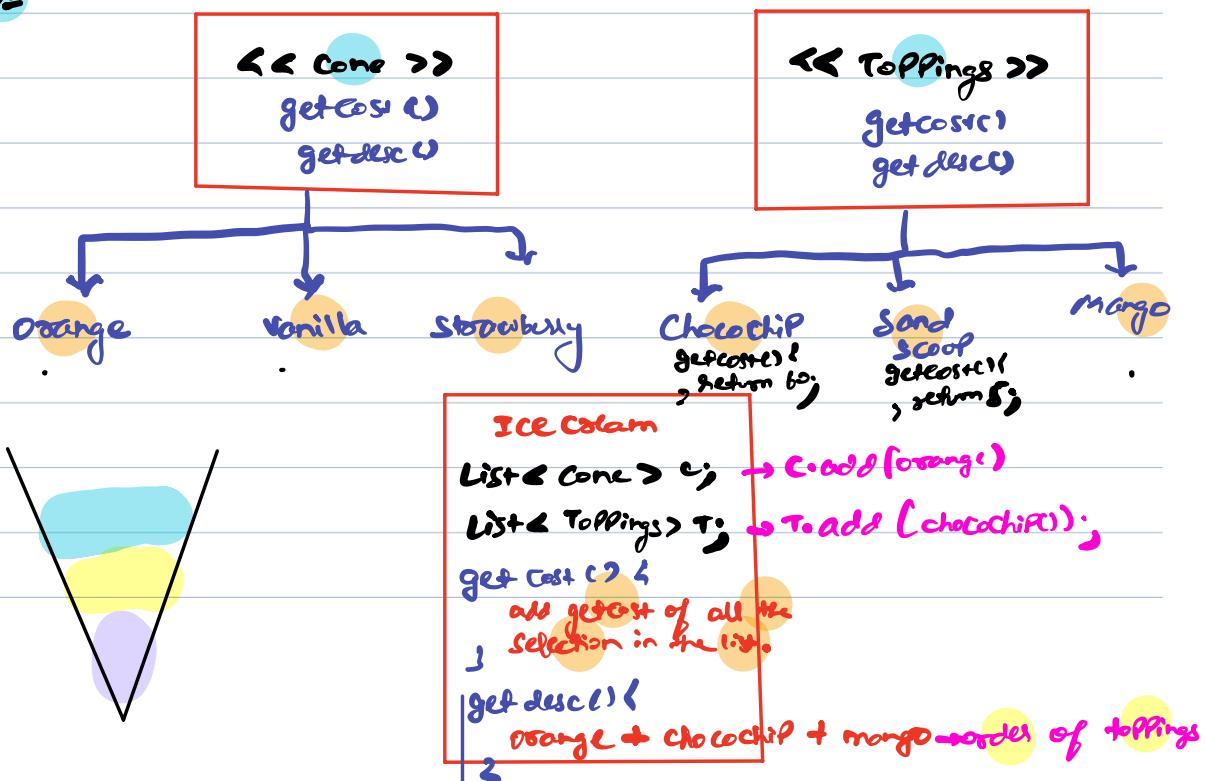


↳ APP can take order of ice-cream
↳ { Cone }
↳ Custom ice-cream
↳ create

→ Show the Cost of ice-cream to user.

→ Description

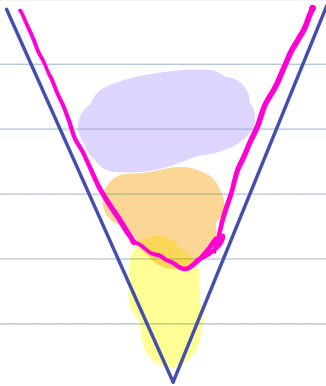
SL





requirement

52



→ selection in order and multiple cones can be selected.

<<Ingredient>>

getcost()

getdesc()

orange

vanilla

strawberry

chocolate chip

chocolate

orange

Ice Cream

List<Ingredient>

getcost()

getdesc()

— — —

orange, chocolate chip, strawberry

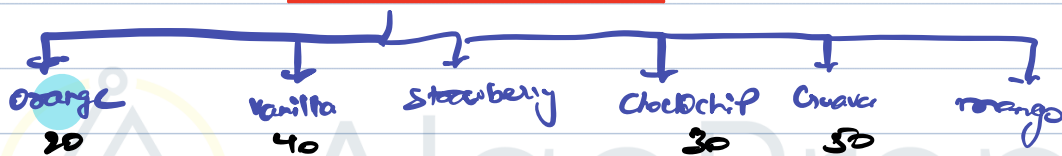
→ in order



↳ When do we need decorators

↳ whenever we need to keep on adding another entity to create your final entity, consider decorators design pattern.

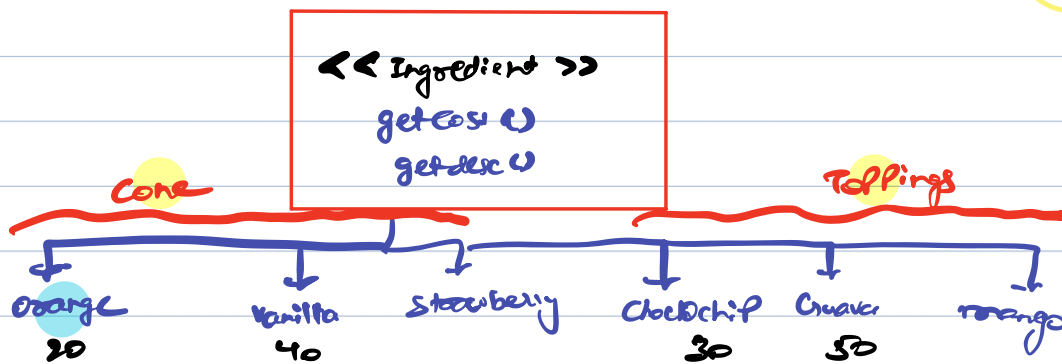
◀◀ Ingredient ▶▶
getCost ()
getDesc ()



Orange cone $\Rightarrow$ getCost () : 20
+

Caramel $\Rightarrow$ getCost () : 20 + Cost of Caramel : 70

Choco chip $\Rightarrow$ getCost () : 70 + Cost of Choco chip : 100



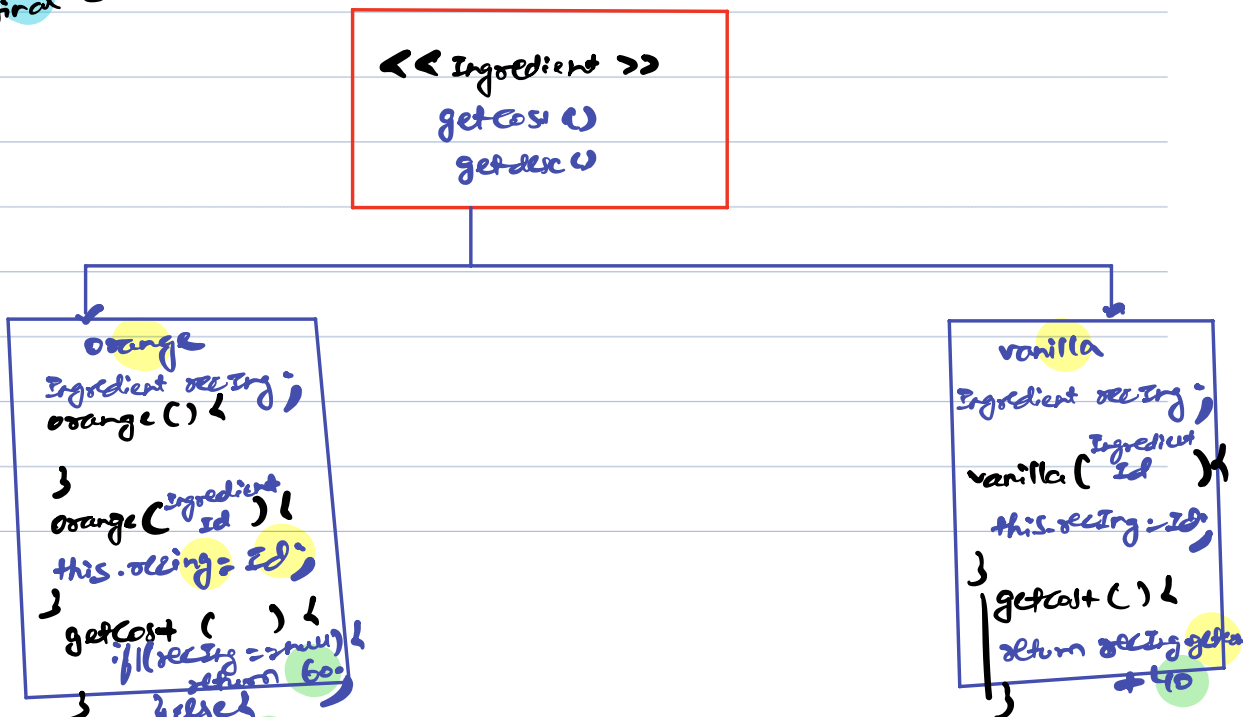
Should act both as base as well as add-on
→ base Selection

↳ orange

Only act as add-on
→ add Ons

chocolate + mango

Final Solⁿ



return ~~self~~ Ing.cost + 60



→ Ingredient IC = new orange (→ 60 + ~~choc.getcost()~~ 150
new choc (← 40 + ~~vanilla.getcost()~~ 110
new vanilla (→ 50 + ~~orange.getcost()~~ 60
new orange ();
),
)

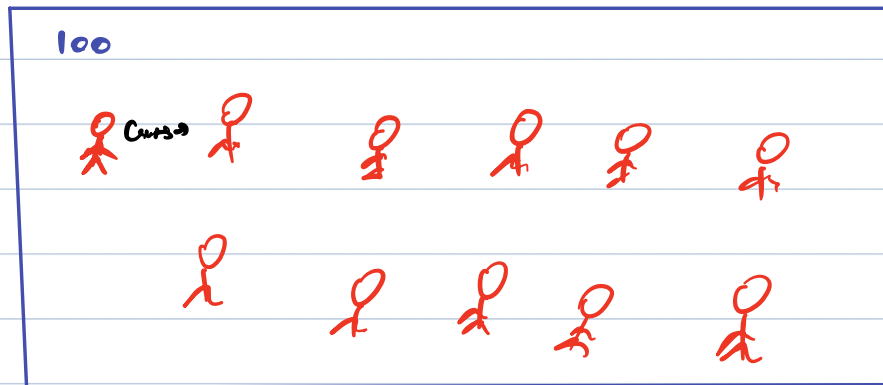
IC.getcost() → 210

↳ code on your own

Break till 9:20 Pm



→ flyweight design Pattern



↳ Start game: → game map is going to be loaded.

↳ Change happening must be broadcasted to the Players.

Players

Guards

visible area to us

vehicles

bullets



Bullets		
→ size	→	8B
→ radius	→	4B
→ damage	→	8B
→ image	→	10 Kb
→ location	→	4B
→ curSpeed	→	8B

→ 100 Person



500 rounds - 1500 rounds



150000 bullets per gun.

assume ↓

100000 bullets per gun.



100000 * 10 Kb

= 10^6 Kb = 16B



AlgoPrep

Flyweight design Pattern

↳ objects have 2 types of properties:

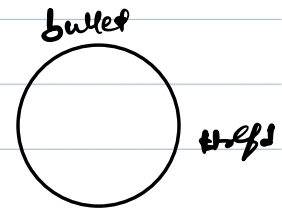
(i) intrinsic : that won't change

(ii) extrinsic : the properties that might change



Split your class into 2 with intrinsic & extrinsic property respect.

Bullets		
→ size	→	8B
→ radius	→	4B
→ damage	→	8B
→ Image	→	10 KB
→ location	→	4B
→ curSpeed	→	8B



contains → flying bullet

→ location	→	4B
→ curSpeed	→	4B
→ bullet b;	→	refs → 4B

Intrinsic → bullet

→ size	→	4B
→ radius	→	4B
→ damage	→	4B
→ Image	→	10 KB

↳ 1000000 * 12B around
 12 * 1000 + 100 → 1 MB

↳ 10KB = 1MB

↳ extending will have some issue as above.